

Acoustic Garden: Exploring Accessibility and Interactive Music with Distance-related Audio Effect Modulation in XR

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ABSTRACT

In AR/XR design, spatial audio and audio-driven narratives are generally considered as secondary to visual interface and storytelling. Non-visual content production and research for visually impaired users are underrepresented. Similarly, in the field of music, visual interfaces dominate discussions on gesture-based sound synthesis, leaving intuitive, audience-driven experiences unexplored. Historically, studies in architecture and music focused on spatial and musical sequences, but they largely remained confined to print media representations.

In this project, inspired by the progressive structure of electronic music, we introduce a spatialized sound synthesis method based on distance-related audio effect modulation combined with binaural spatialization. This approach is intended to navigate users through space without depending on visual indicators, utilizing auditory cues from a multitude of virtual audio objects instead. These objects are responsive to user movements, providing dynamic and immersive musical experiences on a walking scale. Interaction with specific audio objects enables users to dictate different musical progressions, which follows a self-similar spatial structure.

Unlike traditional sonification methods that translate data into plain audio, our approach emphasizes the emotional impact of auditory messages. It unveils the potential for audience-involved, spatially-driven musical narratives. Testing across various platforms, we faced challenges in sound design, hardware limitations, and cognitive processing.

KEYWORDS

Mixed Reality, Augmented Reality, Spatial Computing, Sonification, Spatial Audio, Accessibility, Interactive Music

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1 INTRODUCTION

Based on the concept of exploring "invisible" acoustic space, the project began with a reflection on sensory hierarchy in art and design, revealing cultural differences between what is considered "visually appealing" versus "acoustically appealing," a dichotomy that perpetuates notions of inequality. The scope was later narrowed to employ augmented reality and spatial audio to investigate the relationship between sound and bodily movement in space. How does sound influence movement, and how does movement generate sound narratives?

These questions inspired "Acoustic Garden": an interactive spatial audio experience in mixed reality, emphasizing auditory exploration at the pedestrian level. Using game engine and augmented reality technology, the project re-imagines electronic music through spatial-data-driven audio effect modulation and binaural audio spatialization. It aims to create a navigational tool that guides users through space using auditory cues from virtual audio objects, rather than visual indicators. This approach places mainstream users in the same position as visually impaired individuals, separating audio narratives from visual ones while connecting both senses through user presence. The goal is not to prioritize one sense over another but to provide an independent option for users who prefer audio. These objects respond to user movement, forming immersive musical experiences and allowing users to shape their own musical narratives through dynamic interaction with surrounding sounds.

Technically, "Acoustic Garden" translates classical electronic music structures into a spatial context. The interplay of sound loops and effects modulation forms a musical structure that evolves through repetition and variation. This system translates spatial data and user movements into sound modulations (such as the frequency of a filter, the grain length of a granular synthesizer, or the speed of an LFO), enabling exploration of virtual visual contexts through varied audio cues that inform users' relationships to their surroundings. This interaction not only enhances the walking experience

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Figure 1: Screenshots of the visual experience: only a limited range of objects are visible to the user, while audio objects are placed beyond visual range.

but transforms it into a real-time musical journey. This approach explores new practices for spatial design, creating inclusive experiences for visually impaired individuals, and pushes the boundaries of AR/XR by offering spatial experiences independent of visual stimuli. In music, it democratizes performance through non-linear interactive music for non-professional users, imagining accessible music via minimal control.

“Acoustic Garden” builds on traditional sonification studies to explore new musical possibilities. It emphasizes the emotional resonance of auditory messages, focusing on audience-involved, spatially driven, non-linear musical narratives. Using ARCore and Unreal Engine Metasounds, we developed an Android mobile platform solution. We implemented spatially mapped data as a wavetable (reflecting a spatial distribution or distance field of audio objects). The user’s location is translated into data streams by sampling the wavetable, and their movements modulate audio attributes. Scenes with hidden visual objects rely on spatial audio cues beyond the visual range, translating movement into audio cues that produce a musical experience. Rectangular fields divide the scene into zones, each with its own musical mechanisms—varying BPM, audio composition, signal processing for audio objects, and play area sizes. Guided by multiple audio cues, old scenes stop, and new scenes are revealed as the user crosses scene borders. Similar to a musical performance system, triggering a new scene uses a BPM quantization mechanism, scheduling musical events on beat while the user interacts with audio objects or scene borders. This ensures the musical structure responds automatically to user interactions.

Besides mapped distance fields, we considered other spatial maps, such as sine wave maps along the x, y, or z axis, hand-drawn maps, and user trail maps. However, tester feedback indicated that these maps interfered with the readability of distances to target audio objects. Therefore, for this project, we focused specifically on researching distance maps to investigate target-oriented navigation.

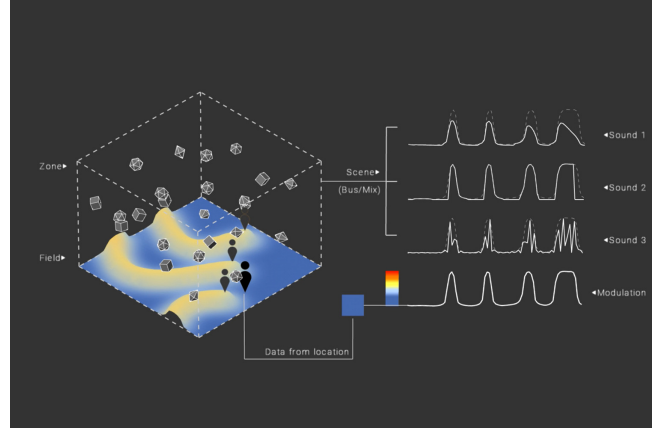


Figure 2: The application takes user’s location to sample a spatial map/wavetable of data and translate into parameters of audio effect modulation.

2 DISCOVERIES

We launched the mobile application test in an open space of a park, where the ground is mostly flat with several trees. Throughout our experiments at pedestrian scale, we found challenges below:

- (1) **Real-World Challenges:** The complexity of the physical world affects user participation in musical experiences. Current devices have limited spatial mapping resolution, making it hard to accurately reflect surroundings. It’s difficult to modify physical obstacles and boundaries unless in a specifically designed environment. Sometimes, audio targets are in inaccessible areas or blocked by objects, especially if not mapped in time. For the next step we aim to develop real-time adaptation of target audio objects and spatial data maps.
- (2) **Device Limitations:** Portable devices offer limited audio spatialization to accurately reflect distance to targets, audio effect modulation partially enhances distance perception.
- (3) **User Input and Feedback:** User inputs are unpredictable and mixed with noise, requiring post-processing to meet musical standards. Quantized data may delay feedback, making user’s spatial situation less readable. Solutions such as joystick might be considered in other contexts such as gaming.
- (4) **Audio Source Distinction:** Untrained users struggle to distinguish directions and variations of multiple audio sources. Even echolocation experts need to turn around to identify audio cues, a process made harder by hardware limitations. Complex audio messages require constant attention and adjustment based on user movements. Further research is needed to improve this mechanism.

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